

Patrick Fanella

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PROFILE

Full-stack / platform engineer building production web systems, AI-enabled workflows, and research-oriented data-management automation. Experienced across Python, SQL, React, Go, TypeScript, PostgreSQL, Docker, Kubernetes, Git, CI/CD, REST APIs, Linux CLI administration, observability, and accessible workflow interfaces. Strong fit for research data workflow development involving data discovery, filtering, sanitization, normalization, AI-enabling metadata, HDF5-oriented packaging, inventory/migration workflows, long-term storage, retrieval, documentation, and staff-facing tools.

CORE TECHNOLOGIES

Languages: Python, Go, TypeScript, SQL

AI/LLM: AI workflow design, LLM orchestration, prompt engineering, structured workflows, RAG / retrieval-augmented generation, retrieval evaluation, tool/function calling, structured outputs, embeddings, semantic indexing, vector search, pgvector, Whisper, speech-to-text

Research Data Workflows: data discovery, ingestion, filtering, sanitization, normalization, metadata design, AI-enabling metadata, inventory workflows, migration automation, HDF5-oriented packaging, long-term storage, retrieval workflows, documentation

Backend/Data: PostgreSQL, Redis, REST APIs, relational databases, async jobs, task queues, idempotency, retries, data modeling, migrations, reconciliation jobs, state machines

Platform/Ops: Linux CLI, Debian, RHEL, Docker, Kubernetes, Nginx, Git, GitHub Actions, CI/CD, Vault, secrets management, Prometheus, Grafana, observability, alerting, runbooks

Frontend/GUI: React, Tailwind CSS, Playwright, accessible web interfaces, workflow UI development, accessibility testing, visual regression testing

Architecture/Security: event-driven systems, background workers, integration architecture, human-in-the-loop workflows, privacy-by-design, security automation

EXPERIENCE

Subcult.tv

Principal Engineer

Chicago

2023-Present

- Built and operated multi-service platforms using React, Go, TypeScript, Python, PostgreSQL, Redis, Docker, AT Protocol, and containerized deployment patterns, supporting data ingestion, discovery, creator workflows, real-time collaboration, and privacy-first user experiences.
- Designed AI-enabled workflows using RAG-style retrieval, pgvector search, embeddings, Whisper transcription, structured outputs, tool/function calling, prompt workflows, agent memory, and multi-agent orchestration.
- Implemented data-processing and workflow automation systems using async jobs, task queues, retry logic, idempotent operations, reconciliation jobs, stateful workflows, and human-in-the-loop review paths for ingestion, moderation, automation, and content preparation.
- Built backend patterns for data inventory, metadata enrichment, search, retrieval, migration, and long-term access across multiple deployed services, aligning product data workflows with structured storage, documentation, and reuse requirements.
- Standardized platform infrastructure with Docker/Kubernetes deployment patterns, Vault-backed secrets, GitHub Actions CI/CD, automated migrations, Prometheus/Grafana monitoring, alerting, and runbooks.
- Added automated quality gates across performance, accessibility, visual regression, load testing, regression detection, privacy controls, EXIF stripping, consent-based telemetry, and security checks.

DeliverZero

Full-Stack Engineer

Chicago/NYC/Denver

2022-2023

- Led tracking, returns, inventory, and dashboard features supporting launch and scale across 3 markets through delivery platform, POS, REST API, and partner integrations.
- Defined integration standards for idempotency, retries, observability, partner test harnesses, lifecycle state models, reconciliation jobs, and failure handling across high-volume operational workflows.
- Built replenishment monitoring, exception routing, self-serve partner workflows, reporting systems, and dashboards, reducing manual operational work by approximately 25%.
- Developed backend data models and reconciliation jobs processing thousands of weekly inventory, container lifecycle, customer-status, POS, delivery, and internal system updates.